

The search continues: Integrating parking into an agent-based transport simulation

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1 INTRODUCTION

The issue of parking has been a long-standing challenge in the field of transport simulation frameworks (Thomas, 2024). Following Rybczak et al. (2024), three different approaches to modeling parking in agent-based simulations can be distinguished: (i) **Explicit simulation of parking search behavior:** For example, Bischoff & Nagel (2017) use the “taxi” implementation of Multi-Agent Transport Simulation (MATSim) to model explicit parking search and demonstrate its applicability. However, they note that a full city-wide implementation is computationally too expensive. Tchervenkov (2022) extends this approach within the iterative MATSim loop, using a 0.1% demand sample to reduce computational cost. (ii) **A global view to parking behavior:** Waraich & Axhausen (2012) propose a parking choice model in MATSim, where agents select among nearby parking options based on generalized parking disutility, including parking cost, walking effort, and additional driving distance. Senozon’s parking proxy (Kohl, 2024) also avoids explicit search by extending egress walking time based on spatial-temporal parking demand from the previous iteration. (iii) **A parameterization of parking pressure:** Schlenther et al. (2022) and Rybczak et al. (2026) apply simplified generalized cost or time penalties to represent parking search. Similarly, Rieser & Bischoff (2024) integrate parking costs based on link attributes in the simulation.

The approaches of explicitly modeling parking search are computationally too expensive. Waraich & Axhausen (2012) and Kohl (2024) are closer to what would be suitable. However, in the paper (Waraich & Axhausen, 2012) it is stated that the parking search time is not included and the focus rather lies on parking choice; additionally, the code is rather old and does not benefit from newer MATSim coding infrastructure. The parking proxy modifies the egress leg in the plan before the mobsim and changes it back directly after; this is something that we would like to avoid. Also, it assumes the parking supply per sqkm as constant throughout the area of simulation, which is elegant, but does not correspond to typical parking data, which gives parking spaces per link (Berlin.de, 2025), and which we would like to be able to use in the long run. The parameterization of parking pressure in the form of time or generalized cost penalties is computationally efficient. However, in the existing implementations these are fixed attributes that do not take into account the interaction of demand and supply, that determine parking search times. Therefore, we propose a new approach to account for parking search times within MATSim.

2 METHODOLOGY

In MATSim, car travel between two activities involves an access walk to the network, a vehicular trip along a predefined route, and an egress walk at the destination. Parking is typically assumed to occur without delay, and parking search or related disutility is not represented.

To integrate parking dynamics into the simulation, we design a modular approach consisting of four main components: (i) **Parking occupancy observer:** This component

tracks the number of parked vehicles within a certain distance of the arrival link. In the initial iteration at the beginning of the simulation, occupancy is derived directly from the initial demand. (ii) **Dynamic parking handler:** This module determines, upon vehicle arrival, whether parking is required for the respective vehicle type (e.g., private cars versus public transport vehicles). (iii) **Parking search time calculator:** This component estimates parking search time by identifying nearby links around the destination and assigning weights based on spatial proximity. It combines local parking occupancy with a search-time function to compute the expected delay. (iv) **Parking search time function:** The search time is calculated using a flexible functional form. In this work, we adopt the formulation of (Belloche, 2015):

$$t_{\text{parking}} = \alpha \cdot \exp(-\beta \cdot T_c) \quad (1)$$

where T_c denotes the congestion ratio, defined as the number of parked vehicles divided by the total parking capacity within a district. The parameters α and β can be taken from Belloche (2015) as we do for our experiments (see (Belloche, 2015) Table 4) or calibrated to the specific scenario. To compute T_c , a weighted kernel function is applied to aggregate parking occupancy and capacity across nearby links. Based on the observed occupancy and available capacity within a discretized spatio-temporal context, an individual parking search time is assigned using Equation 1. Agents leave their vehicles and perform their egress leg once the waiting time has elapsed. Thus, we do not simulate parking search physically on the network.

3 RESULTS

Consider the grid network, with 324 agents following a home–work–shopping–home pattern. The simulation is executed without replanning. Each link is assigned 10 parking spaces (1,800 in total), with parameters $\alpha = 0.11$ and $\beta = -8.586$ in Equation 1.

Under these conditions, total travel time increases slightly from 3,223 to 3,224 h. Reducing supply to five spaces per link (900 total) raises travel time to 3,261 h. Figure 1 shows the corresponding distributions, where parking search times increase markedly under scarce supply.

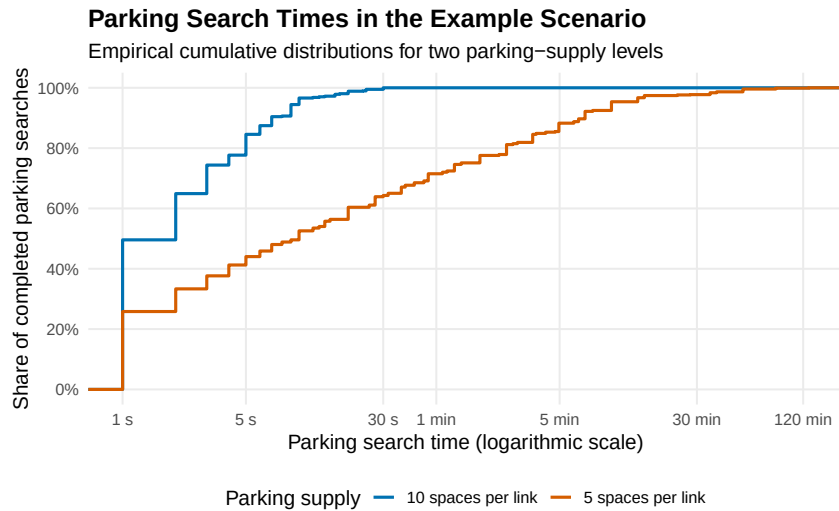


Figure 1: Empirical cumulative distributions of parking search times in the illustrative grid scenario. The horizontal axis uses a logarithmic scale.

4 CONCLUSION

We introduced a new approach to represent parking search within MATSim. The framework enables the modeling of parking-induced delays by capturing both spatial heterogeneity and temporal waiting dynamics. In combination with existing approaches for parking choice and parking cost, this contributes to a more comprehensive integration of parking into the simulation framework and supports the analysis of parking-related policies.

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REFERENCES

- Belloche, S. (2015). On-street parking search time modelling and validation with survey-based data. *Transportation Research Procedia*, 6, 313–324. doi: 10.1016/j.trpro.2015.03.024
- Berlin.de. (2025). *1,2 Millionen Parkplätze: Neue Karte zeigt Details*. Retrieved from <https://www.berlin.de/aktuelles/9275007-958090-12-millionen-parkplaetze-neue-karte-zeig.html> (Accessed: 2025-02-28)
- Bischoff, J., & Nagel, K. (2017). Integrating explicit parking search into a transport simulation. *Procedia Computer Science*, 109, 881–886. Retrieved from <http://www.sciencedirect.com/science/article/pii/S1877050917310906> doi: 10.1016/j.procs.2017.05.414
- Kohl, T. (2024). *Parkingproxy*. https://www.matsim.org/conferences/mum2024/presentations/MUM2024_Kohl_ParkingProxy.pdf.
- Rieser, M., & Bischoff, J. (2024). *MATSim parking costs module*. <https://github.com/matsim-org/matsim-libs/tree/main/contribs/parking/src/main/java/org/matsim/contrib/parking/parkingcost>. (Accessed: 2026-04-15)
- Rybczak, G., Heinrich, P., Leich, G., Rakow, C., Ewert, R., Gast, K., ... Nagel, K. (2026). *Calibration and validation of large-scale agent-based transport models*. (Submitted to the Euro Working Group on Transportation Annual Meeting 2026)
- Rybczak, G., Meinhardt, S., Schlenther, T., Rakow, C., Ziemke, D., & Nagel, K. (2024). Agent-based modeling of residential parking zones in Leipzig. *Procedia Computer Science*, 238, 749–756. doi: 10.1016/j.procs.2024.06.087
- Schlenther, T., Wagner, P., Rybczak, G., Nagel, K., Bieker-Walz, L., & Ortgiess, M. (2022). Simulation-based investigation of transport scenarios for Hamburg. *Procedia Computer Science*, 201, 587–593. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1877050922004896> doi: 10.1016/j.procs.2022.03.076
- Tchervenkova, C. (2022). *Empirical and simulation studies on parking in Switzerland* (Doctoral Thesis, ETH Zurich, Zurich). doi: 10.3929/ethz-b-000593365
- Thomas, A. (2024). *Exploration of traffic area segmentation on aerial imagery to address the parking data requirements of travel demand models*. Retrieved from https://www.mos.ed.tum.de/fileadmin/w00ccp/tb/theses/2024/Thomas_2024_webVersion.pdf
- Waraich, R., & Axhausen, K. W. (2012, 12). Agent-based parking choice model. *Transportation Research Record Journal of the Transportation Research Board*, 2319, 39–46. doi: 10.3141/2319-05